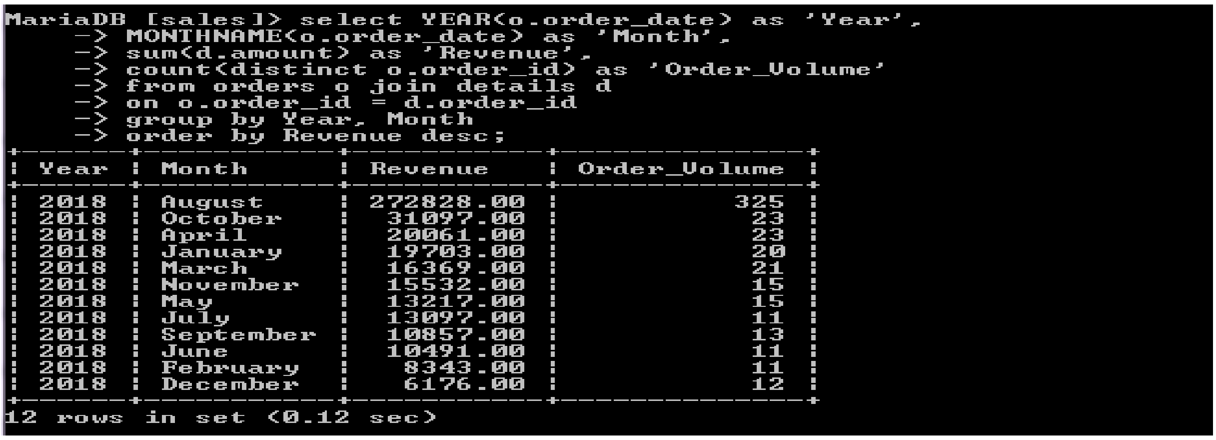
TASK 6: Sales Trend Analysis Using Aggregations

Objective: Analyze monthly revenue and order volume.
Tools: MySQL

1. **Monthly Revenue and Order Volume**

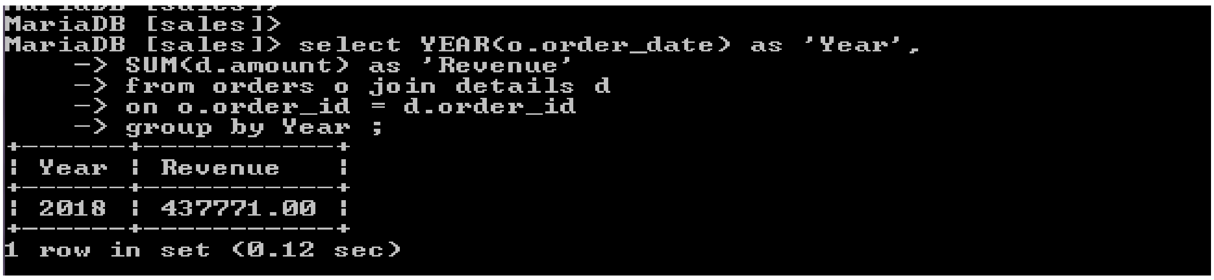
   select YEAR(o.order_date) as 'Year', MONTHNAME(o.order_date) as 'Month', sum(d.amount) as 'Revenue', count(distinct o.order_id) as 'Order_Volume' from orders o join details d on o.order_id = d.order_id group by Year, Month order by Revenue desc;



```
MariaDB [sales]> select YEAR(o.order_date) as 'Year',
    -> MONTHNAME(o.order_date) as 'Month',
    -> sum(d.amount) as 'Revenue',
    -> count(distinct o.order_id) as 'Order_Volume'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> group by Year, Month
    -> order by Revenue desc;
+------+-----------+-----------+--------------+
| Year | Month     | Revenue   | Order_Volume |
+------+-----------+-----------+--------------+
| 2018 | August    | 272828.00 |          325 |
| 2018 | October   |  31097.00 |           23 |
| 2018 | April     |  20061.00 |           23 |
| 2018 | January   |  19703.00 |           20 |
| 2018 | March     |  16369.00 |           21 |
| 2018 | November  |  15532.00 |           15 |
| 2018 | May       |  13217.00 |           15 |
| 2018 | July      |  13097.00 |           11 |
| 2018 | September |  10857.00 |           13 |
| 2018 | June      |  10491.00 |           11 |
| 2018 | February  |   8343.00 |           11 |
| 2018 | December  |   6176.00 |           12 |
+------+-----------+-----------+--------------+
12 rows in set (0.12 sec)
```

2. **Revenue by Year**

   select YEAR(o.order_date) as 'Year', SUM(d.amount) as 'Revenue' from orders o join details d on o.order_id = d.order_id group by Year ;



```
MariaDB [sales]>
MariaDB [sales]> select YEAR(o.order_date) as 'Year',
    -> SUM(d.amount) as 'Revenue'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> group by Year ;
+------+-----------+
| Year | Revenue   |
+------+-----------+
| 2018 | 437771.00 |
+------+-----------+
1 row in set (0.12 sec)
```

3. **Order Volume by Month**

   select MONTHNAME(o.order_date) as 'Month', COUNT(distinct o.order_id) as 'Order_Volume' from orders o join details d on o.order_id = d.order_id group by Month order by Order_Volume desc;

```
MariaDB [sales]>
MariaDB [sales]> select MONTHNAME(o.order_date) as 'Month',
    -> COUNT(distinct o.order_id) as 'Order_Volume'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> group by Month order by Order_Volume desc;
+-----------+--------------+
| Month     | Order_Volume |
+-----------+--------------+
| August    |          325 |
| April     |           23 |
| October   |           23 |
| March     |           21 |
| January   |           20 |
| May       |           15 |
| November  |           15 |
| September |           13 |
| December  |           12 |
| February  |           11 |
| July      |           11 |
| June      |           11 |
+-----------+--------------+
12 rows in set (0.13 sec)
```

4. **Monthly Profit**

select MONTHNAME(o.order_date) as 'Month', SUM(d.profit) as 'Profit' from orders o join details d on o.order_id = d.order_id group by Month order by Profit desc;

```
MariaDB [sales]> select MONTHNAME(o.order_date) as 'Month',
    -> SUM(d.profit) as 'Profit'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> group by Month
    -> order by Profit desc;
+-----------+----------+
| Month     | Profit   |
+-----------+----------+
| August    | 71729.00 |
| January   |  7416.00 |
| October   |  5176.00 |
| April     |  4694.00 |
| November  |  4660.00 |
| May       |  3865.00 |
| March     |  3063.00 |
| September |  3003.00 |
| June      |  2792.00 |
| July      |  2705.00 |
| February  |  2277.00 |
| December  |  1741.00 |
+-----------+----------+
12 rows in set (0.16 sec)
```

5. **Revenue for a Specific Month**

select MONTHNAME(o.order_date) as 'Month', SUM(d.amount) as 'Revenue' from orders o join details d on o.order_id = d.order_idwhere MONTHNAME(o.order_date) = 'August' group by Month order by Month;

```
MariaDB [sales]>
MariaDB [sales]> select MONTHNAME(o.order_date) as 'Month',
    -> SUM(d.amount) as 'Revenue'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> where MONTHNAME(o.order_date) = 'August'
    -> group by Month
    -> order by Month;
+--------+-----------+
| Month  | Revenue   |
+--------+-----------+
| August | 272828.00 |
+--------+-----------+
1 row in set (0.08 sec)
```

6. **Lowest Revenue Month using EXTRA() and LIMIT**

   select EXTRACT(MONTH FROM o.order_date) as 'Month', MONTHNAME(o.order_date) as 'Month_Name', SUM(d.amount) as 'Revenue' from orders o join details d on o.order_id = d.order_id group by Month order by Revenue limit 1;

```
MariaDB [sales]>
MariaDB [sales]> select EXTRACT(MONTH FROM o.order_date) as 'Month',
    -> MONTHNAME(o.order_date) as 'Month_Name',
    -> SUM(d.amount) as 'Revenue'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> group by Month order by Revenue limit 1;
+-------+------------+---------+
| Month | Month_Name | Revenue |
+-------+------------+---------+
|    12 | December   | 6176.00 |
+-------+------------+---------+
1 row in set (0.11 sec)
```

7. **Highest Revenue Month using EXTRA() and LIMIT**

   select EXTRACT(MONTH FROM o.order_date) as 'Month', MONTHNAME(o.order_date) as 'Month_Name', SUM(d.amount) as 'Revenue' from orders o join details d on o.order_id = d.order_id group by Month order by Revenue desc limit 1;

```
MariaDB [sales]>
MariaDB [sales]> select EXTRACT(MONTH FROM o.order_date) as 'Month',
    -> MONTHNAME(o.order_date) as 'Month_Name',
    -> SUM(d.amount) as 'Revenue'
    -> from orders o join details d
    -> on o.order_id = d.order_id
    -> group by Month order by Revenue desc limit 1;
+-------+------------+-----------+
| Month | Month_Name | Revenue   |
+-------+------------+-----------+
|     8 | August     | 272828.00 |
+-------+------------+-----------+
1 row in set (0.12 sec)
```